

As the Agile Project Manager for the **AI-Powered Hospital Management System**, I have structured this roadmap using a 24-week (6-month) timeline, divided into 2-week Sprints. We will follow a Scrum framework, prioritizing core clinical functionality followed by AI integration and compliance.

1. Estimated Duration

- **Total Duration:** 24 Weeks (6 Months)
- **Development Model:** Agile/Scrum (12 Sprints)

2. Milestones

- M1 (Week 4):** Foundation & Infrastructure (HIPAA/Security Architecture).
- M2 (Week 10):** Core HMS Module (Patient Mgmt, Scheduling, EHR).
- M3 (Week 16):** AI Integration (Triage & Clinical Decision Support).
- M4 (Week 20):** Reporting, Billing, and Resource Management.
- M5 (Week 24):** Final UAT, Regulatory Audit, and Deployment.

3. Weekly Timeline

Phase	Weeks	Focus
	1-2	Discovery
	3-4	Infrastructure
	5-8	HMS Core
	9-12	AI Alpha
	13-16	Clinical Suite
	17-20	Admin & Billing
	21-22	Hardening
	23-24	Release

****4. Key Deliverables****

- ****Technical:**** HIPAA-compliant cloud architecture, encrypted EHR database, API middleware for imaging systems.
- ****Product:**** Web-based portal with role-based dashboards (Admin, Doctor, Patient).
- ****AI:**** Trained ML models for Triage (90% accuracy) and Clinical Summary engine.
- ****Documentation:**** Technical System Documentation, User Manuals, Data Privacy Compliance Report.

****5. Risk Assessment & Mitigation****

Risk Impact Mitigation Strategy		
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Data Privacy Breach	Critical	Implement zero-trust architecture, multi-factor authentication (MFA), and automated audit logs.
AI Hallucinations	High	Mandatory "Human-in-the-loop" verification workflow; AI suggestions strictly advisory, not prescriptive.
User Resistance	Medium	Early involvement of clinical "champions" in UI/UX design; focus on "time-saving" features rather than "replacing" staff.
Regulatory Hurdles	High	Engage legal/compliance officers during Sprint 1; ensure documentation mirrors local regulatory standards.

****Agile Best Practices to Implement:****

- ****Sprint Reviews:**** Demo functional increments to clinical stakeholders every 2 weeks to ensure UI/UX fits a medical environment.
- ****Retrospectives:**** Evaluate the performance of the AI models during every sprint; adjust training data sets based on physician feedback.
- ****Backlog Refinement:**** The Product Owner will prioritize the "Patient Triage" and "EHR" features to ensure core safety functions are ready for testing as early as possible.
- ****Definition of Done (DoD):**** A task is only "Done" when it passes unit tests, code review, and HIPAA-compliance security checks.